

2026 Operating System

Lab3. UTM 실행 시 오류 안내 (MacOS)

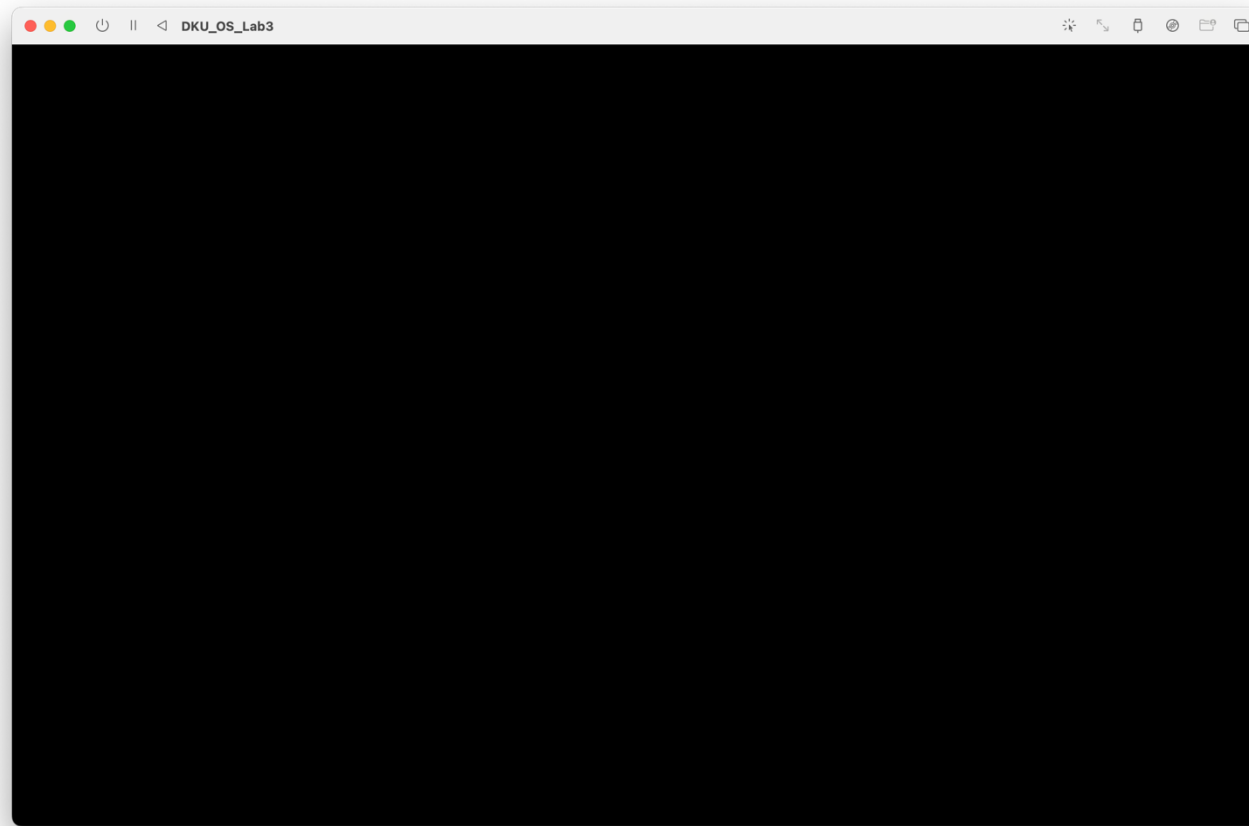
2026.06.01

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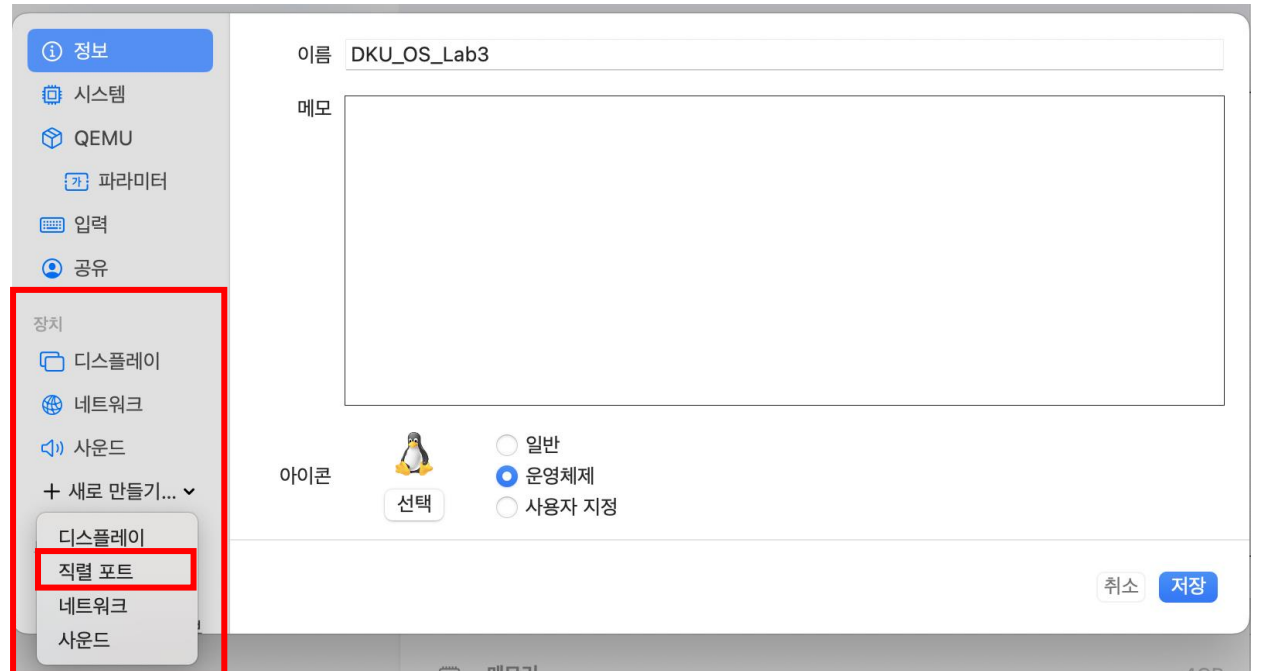
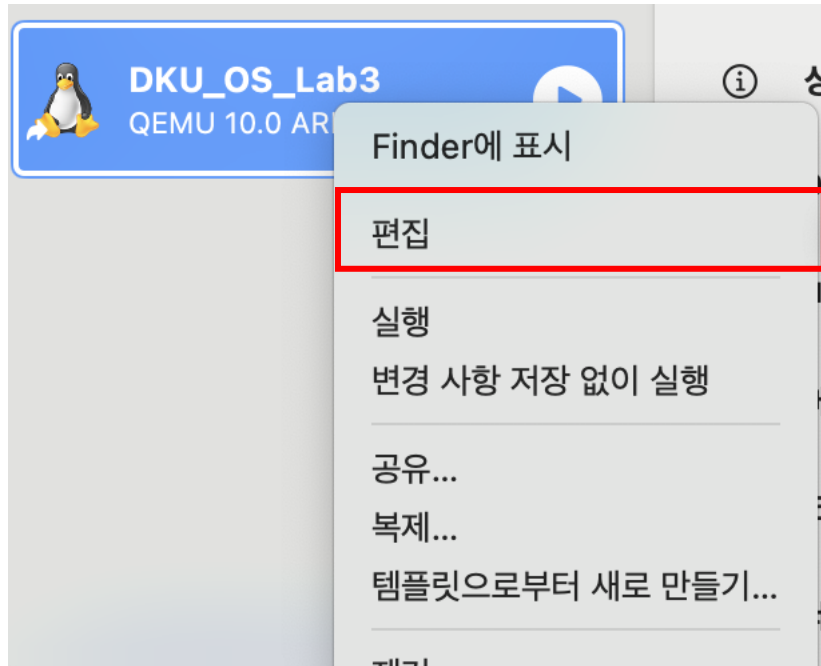
UTM 초기 실행 시 오류 화면

- 디스플레이 출력이 없고, 검정 화면만 뜸



해결 방안

- 가상머신 종료 후, 우클릭을 하여 '편집'에 들어감
- 좌측 메뉴 중 '장치' 에서 '새로 만들기' → '직렬 포트' 선택



해결 방안

- 직렬 포트 선택 시, '모드' → '내장 터미널'
- 저장 후 실행

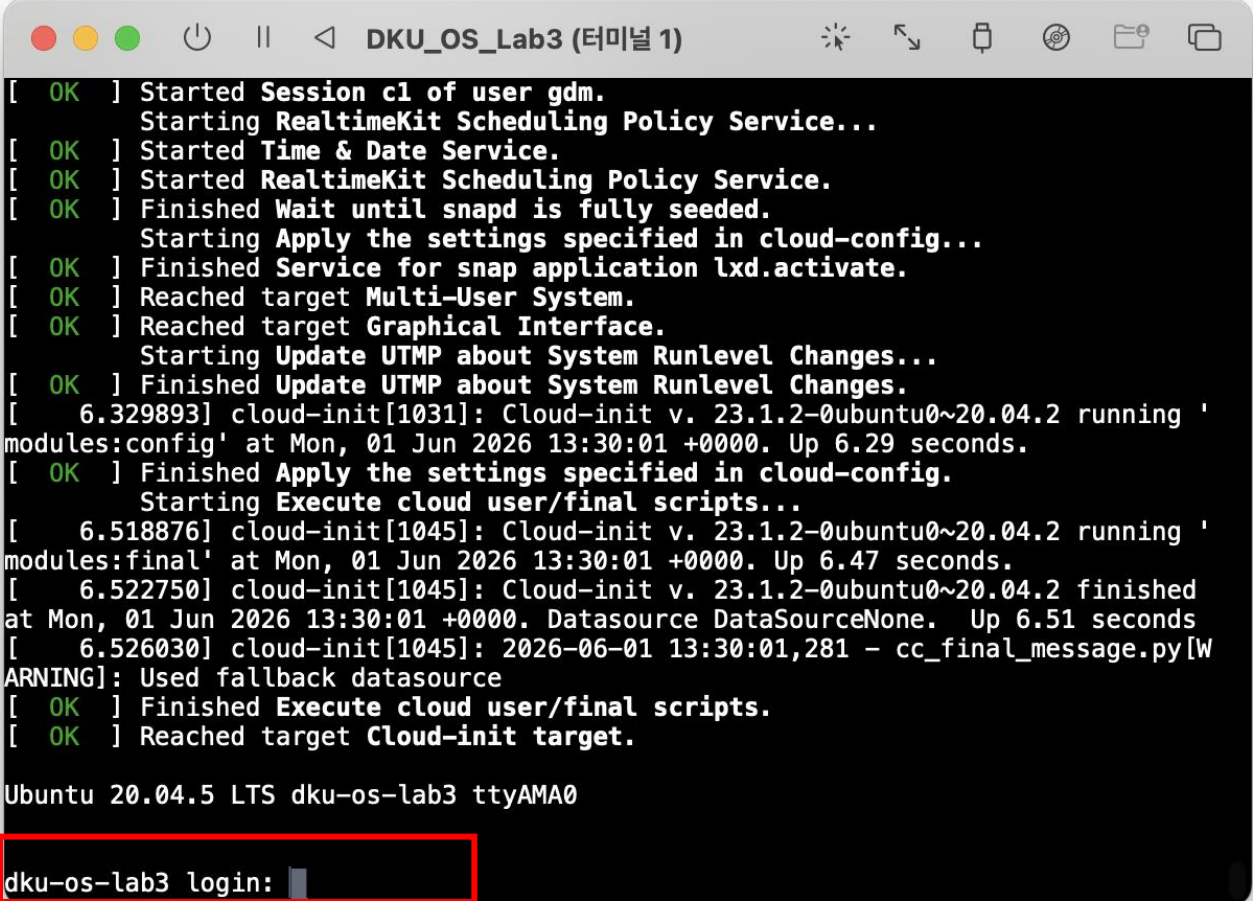
The screenshot shows the QEMU configuration window. On the left is a sidebar with various settings categories. The 'Serial Port' (직렬 포트) option is selected and highlighted in blue. The main area displays the configuration for the serial port, with the 'Mode' (모드) dropdown menu set to 'Built-in terminal' (내장 터미널). Other settings like 'Target' (대상), 'Style' (스타일), 'Theme' (테마), 'Text Color' (글자 색상), 'Background Color' (배경 색상), 'Font' (글꼴), and 'Font Size' (글꼴 크기) are also visible. At the bottom right, the 'Save' (저장) button is highlighted with a red box, indicating the next step in the process.

정보
시스템
QEMU
가 파라미터
입력
공유
장치
디스플레이
직렬 포트
네트워크
사운드
+ 새로 만들기...
드라이브
USB 드라이브

연결
모드 내장 터미널
대상 자동 직렬 장치 (최대 4개)
스타일
테마 기본값
글자 색상
배경 색상
글꼴
글꼴 크기 12
☒ 커서 깜빡이기
콘솔 크기 조정 명령어
stty cols \$COLS rows \$ROWS
제거 취소 저장

해결 방안

- 다음과 같이 터미널이 출력됨
- dku-os-lab3 login: **user**
- password: **0000**



```
[ OK ] Started Session c1 of user gdm.
Starting RealtimeKit Scheduling Policy Service...
[ OK ] Started Time & Date Service.
[ OK ] Started RealtimeKit Scheduling Policy Service.
[ OK ] Finished Wait until snapd is fully seeded.
Starting Apply the settings specified in cloud-config...
[ OK ] Finished Service for snap application lxd.activate.
[ OK ] Reached target Multi-User System.
[ OK ] Reached target Graphical Interface.
Starting Update UTMP about System Runlevel Changes...
[ OK ] Finished Update UTMP about System Runlevel Changes.
[ 6.329893] cloud-init[1031]: Cloud-init v. 23.1.2-0ubuntu0~20.04.2 running '
modules:config' at Mon, 01 Jun 2026 13:30:01 +0000. Up 6.29 seconds.
[ OK ] Finished Apply the settings specified in cloud-config.
Starting Execute cloud user/final scripts...
[ 6.518876] cloud-init[1045]: Cloud-init v. 23.1.2-0ubuntu0~20.04.2 running '
modules:final' at Mon, 01 Jun 2026 13:30:01 +0000. Up 6.47 seconds.
[ 6.522750] cloud-init[1045]: Cloud-init v. 23.1.2-0ubuntu0~20.04.2 finished
at Mon, 01 Jun 2026 13:30:01 +0000. Datasource DataSourceNone. Up 6.51 seconds
[ 6.526030] cloud-init[1045]: 2026-06-01 13:30:01,281 - cc_final_message.py[W
ARNING]: Used fallback datasource
[ OK ] Finished Execute cloud user/final scripts.
[ OK ] Reached target Cloud-init target.

Ubuntu 20.04.5 LTS dku-os-lab3 ttyAMA0

dku-os-lab3 login: 
```

해결 방안

- `df -h` 로 현재 디스크 공간을 확인해보면 30G 중 30G를 사용 중임
 - 이 중 약 20G를 차지 중인 `/EMPTY` 삭제
→ `rm -f /EMPTY`
 - 다시 디스크 공간을 확인하면 20G 사용 가능
 - 확인 후 최종적으로 쉘에서 '`sudo reboot`' 실행
-
- 실행 명령어 순서
 - `df -h` → `rm -f /EMPTY` → `df -h` → `sudo reboot`

The image shows a terminal window titled 'DKU_OS_Lab3_2 (터미널 1)'. It displays two 'df -h' commands and their outputs. In the first output, the root filesystem is at 100% usage. In the second output, after deleting /EMPTY, the usage is reduced to 31%.

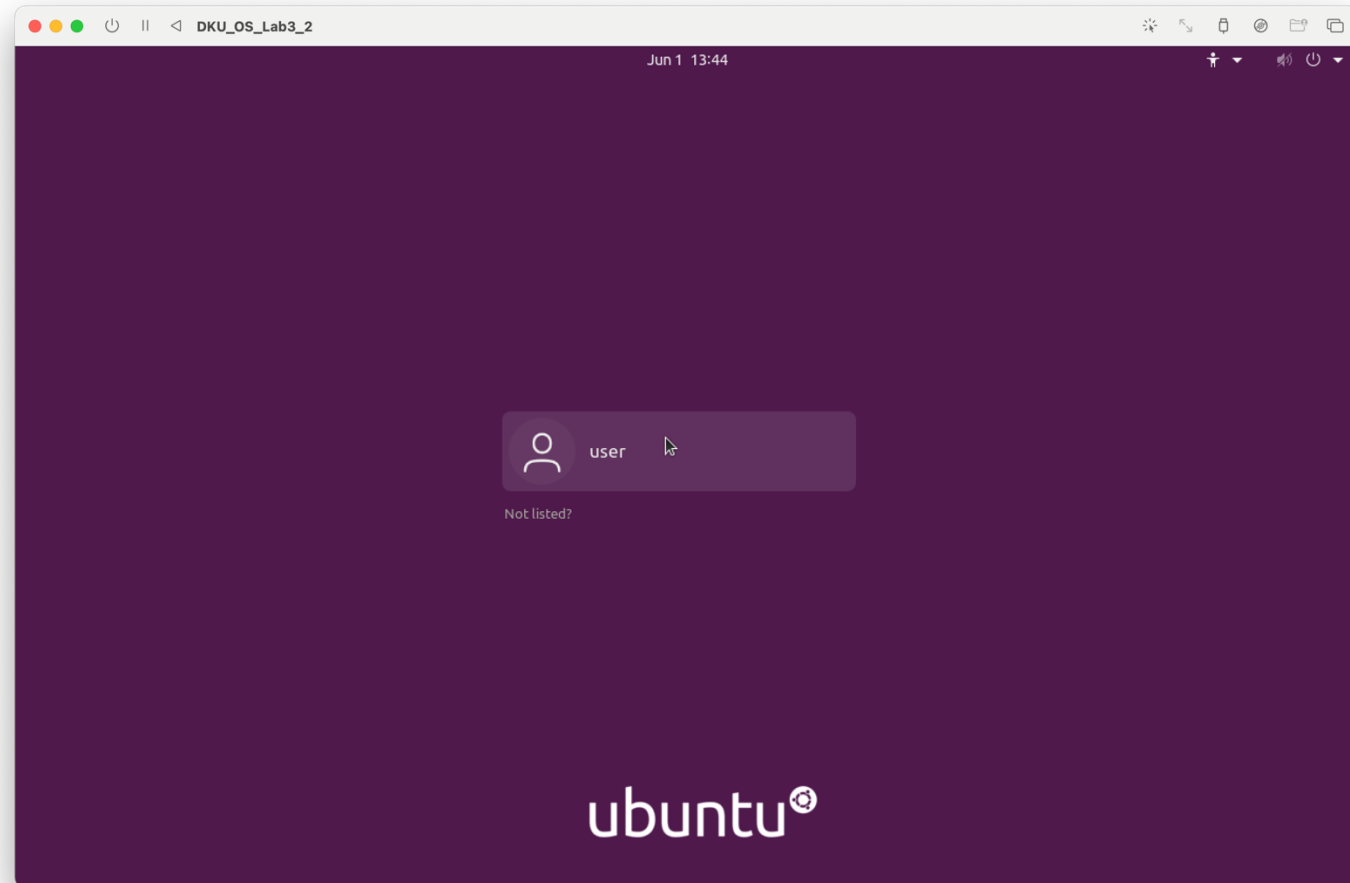
```
user@dku-os-lab3:~$ df -h
Filesystem                Size      Used Avail Use% Mounted on
udev                     1.9G         0   1.9G   0% /dev
tmpfs                     302M       1.4M   301M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 30G       30G    0 100% /
tmpfs                     2.0G         0   2.0G   0% /dev/shm
tmpfs                     5.0M     4.0K   5.0M   1% /run/lock
tmpfs                     2.0G         0   2.0G   0% /sys/fs/cgroup
/dev/vda2                 2.0G     112M   1.7G   7% /boot
/dev/loop0                58M       58M    0 100% /snap/core20/1614
/dev/loop1                62M       62M    0 100% /snap/lxd/22761
/dev/vda1                 1.1G     6.4M   1.1G   1% /boot/efi
/dev/loop2                41M       41M    0 100% /snap/snapd/16299
tmpfs                     392M     4.0K   392M   1% /run/user/1000

user@dku-os-lab3:~$ ls /EMPTY
/EMPTY
user@dku-os-lab3:~$ sudo rm -f /EMPTY
[sudo] password for user:
user@dku-os-lab3:~$ df -h
Filesystem                Size      Used Avail Use% Mounted on
udev                     1.9G         0   1.9G   0% /dev
tmpfs                     302M       1.4M   301M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 30G       8.5G    20G  31% /
tmpfs                     2.0G         0   2.0G   0% /dev/shm
tmpfs                     5.0M     4.0K   5.0M   1% /run/lock
tmpfs                     2.0G         0   2.0G   0% /sys/fs/cgroup
/dev/vda2                 2.0G     112M   1.7G   7% /boot
/dev/loop0                58M       58M    0 100% /snap/core20/1614
/dev/loop1                62M       62M    0 100% /snap/lxd/22761
/dev/vda1                 1.1G     6.4M   1.1G   1% /boot/efi
/dev/loop2                41M       41M    0 100% /snap/snapd/16299
tmpfs                     392M     4.0K   392M   1% /run/user/1000

user@dku-os-lab3:~$
```

해결 방안

- 최종적으로 디스플레이가 잘 출력되는 것을 확인 할 수 있음



Thank you

2026.06.01

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